

Chapter 2: Prompt Engineering Basics

Class 11: Practical / Activities (10 Minutes)

Topic: Compare and Improve AI Prompts

Activity 1: Compare Two AI Prompts (4 Minutes)

Objective

To help students compare weak and well-written AI prompts, understand how prompt quality affects AI responses, and identify the characteristics of an effective prompt.

Materials Required

- Computer/Laptop or Tablet
 - Internet Connection
 - AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, etc.)
 - Prompt Comparison Worksheet
 - Notebook & Pen
 - Teacher Observation Sheet
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Instructions

1. Open the AI tool.
2. Read the two prompts provided by the teacher.
3. Enter **Prompt A** into the AI tool and observe the response.
4. Enter **Prompt B** into the AI tool and observe the response.
5. Compare both responses based on:
 - Accuracy
 - Clarity
 - Organization
 - Level of Detail
 - Suitability for Class IX students

- Record your observations in the worksheet.

Prompt Comparison Example

Prompt A (Weak Prompt)

Tell me about renewable energy.

Prompt B (Improved Prompt)

Explain renewable energy for Class IX students using simple English. Include the definition, five types of renewable energy, advantages, disadvantages, and real-life examples in bullet points.

Prompt Comparison Worksheet

Evaluation Criteria	Prompt A	Prompt B
Clear Topic	☐	☐
Audience Mentioned	☐	☐
Output Format Specified	☐	☐
Detailed Information	☐	☐
Easy to Understand	☐	☐
Better AI Response	☐	☐

Student Activity Checklist

Students should:

- ✓ Test both prompts.
- ✓ Compare AI responses.
- ✓ Identify missing details.
- ✓ Explain which prompt is better.
- ✓ Record observations.

Teacher Observation Checklist

- ✓ Compares prompts correctly.
 - ✓ Identifies differences in AI responses.
 - ✓ Demonstrates analytical thinking.
 - ✓ Participates actively.
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Activity 2: Improve an Existing Prompt (3 Minutes)

Objective

To help students refine a weak AI prompt by adding context, audience, output format, and specific instructions.

Materials Required

- AI Tool
 - Prompt Improvement Worksheet
 - Teacher Evaluation Sheet
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Instructions

1. Choose one weak prompt from the worksheet.
2. Identify the missing information.
3. Rewrite the prompt by adding:
 - Clear topic
 - Target audience
 - Output format
 - Number of points or slides
 - Simple language requirement
4. Generate a new AI response.

5. Compare it with the original response.

Prompt Improvement Examples

Weak Prompt	Improved Prompt
Write about pollution.	Write a 200-word article about air pollution for Class IX students using simple English, headings, and bullet points.
Explain computers.	Explain the basic parts of a computer with labelled descriptions and examples suitable for Class IX students .
Create a presentation.	Create a 5-slide presentation outline on Digital Safety with bullet points, images to include, and a conclusion.

Prompt Improvement Worksheet

Original Prompt	_____
Improved Prompt	_____
What Did You Add?	_____
Why Is It Better?	_____

Discussion Questions

1. Which improvements made the prompt more effective?
 2. How did adding the audience improve the response?
 3. Why is specifying the output format important?
 4. How does prompt refinement save time?
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Teacher Observation Checklist

- ☒ Refines prompts successfully.
- ☒ Uses clear and detailed instructions.
- ☒ Improves AI-generated responses.
- ☒ Explains improvements confidently.

Activity 3: Best Prompt Challenge & Peer Review (3 Minutes)

Objective

To encourage students to write the most effective AI prompt, present it to the class, and receive constructive peer feedback.

Materials Required

- AI Tool
 - Prompt Challenge Worksheet
 - Peer Review Sheet
 - Teacher Observation Sheet
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Instructions

1. Select any educational topic.
 2. Write the best AI prompt possible.
 3. Generate the AI response.
 4. Present your prompt and response to the class.
 5. Classmates review the prompt using the evaluation sheet.
 6. Improve the prompt based on the feedback received.
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Suggested Topics

- Artificial Intelligence
 - Water Conservation
 - Cyber Safety
 - Healthy Lifestyle
 - Space Exploration
 - Climate Change
 - Renewable Energy
 - Digital Citizenship
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Peer Review Worksheet

Evaluation Criteria	Excellent	Good	Needs Improvement
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Prompt Clarity	☐	☐	☐
Context Provided	☐	☐	☐
Output Format	☐	☐	☐
Creativity	☐	☐	☐
AI Response Quality	☐	☐	☐

Student Tasks

Students should:

- ✓ Write an effective AI prompt.
 - ✓ Generate an AI response.
 - ✓ Present their prompt confidently.
 - ✓ Accept peer feedback.
 - ✓ Improve the prompt if needed.
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Teacher Observation Checklist

- ✓ Demonstrates excellent prompt writing.
 - ✓ Explains prompt improvements clearly.
 - ✓ Uses AI responsibly.
 - ✓ Participates confidently in peer review.
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Mini Practical Assessment Rubric

Assessment Area	Marks
Prompt Comparison Skills	20
Prompt Improvement	20
AI Response Evaluation	20
Prompt Presentation & Creativity	20
Participation & Communication	20
Total	100 Marks

Teacher Observation Checklist

- ✓ Compares AI prompts effectively.
 - ✓ Improves prompts by adding clarity, context, and structure.
 - ✓ Evaluates AI-generated responses critically.
 - ✓ Demonstrates creative and responsible prompt engineering.
 - ✓ Participates actively throughout the practical session.
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Practical Time Distribution

Activity	Duration
Compare Two AI Prompts	4 Minutes
Improve an Existing Prompt	3 Minutes
Best Prompt Challenge & Peer Review	3 Minutes
Total Practical Time	10 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Compare weak and well-structured AI prompts and identify the differences in AI-generated responses.
- Improve prompts by adding **clear instructions, context, audience, output format, and specific requirements**.
- Evaluate AI responses based on **accuracy, clarity, organization, relevance, and usefulness**.
- Present and explain their prompt-writing strategy while giving and receiving constructive peer feedback.
- Apply prompt refinement techniques to generate more accurate, meaningful, and high-quality AI outputs.
- Demonstrate responsible, ethical, and effective use of prompt engineering skills for educational, research, and creative tasks.